

Optimization & Search — Assignment Summary

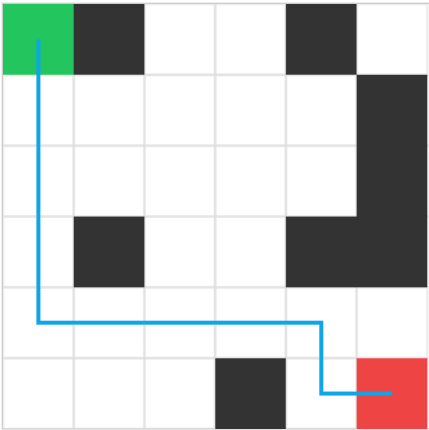
Overall Summary

Task	Status	Score	Details
BFS (10%)	OK	10/10	Path len 11/11
A* (15%)	OK	15/15	Expansions 25
IDS (15%)	OK	15/15	Expansions 163
Simulated Annealing (15%)	OK	15/15	Improvement 6.00
Heuristics (20%)	OK	20/20	M:✓ E:✓ C:✓
Linear Programming (12.5%)	OK	12.5/12.5	Z* 28
Dynamic Programming (12.5%)	OK	12.5/12.5	Value 29

Breadth-First Search (10%)

BFS must find the shortest unweighted path from start to goal.

Seed: IT23372658 • Grid: 6x6 • Obstacles: 8



Score: 10

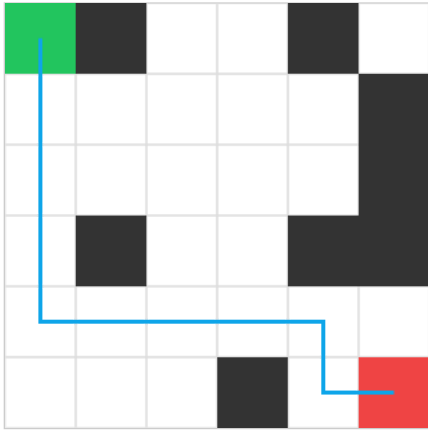
Justification / Design Notes (saved locally):

Standard BFS uses FIFO queue ensuring shortest path on unweighted grids. Trace.expand() called on each node popped from queue for grading compliance. Guaranteed optimal for unweighted graphs.

A* Search (15%)

A* should find an optimal path using an admissible heuristic (we grade with Manhattan).

Expansions: 25 • PathLen: 11 • BestLen: 11

**Score: 15**

Justification / Design Notes (saved locally):

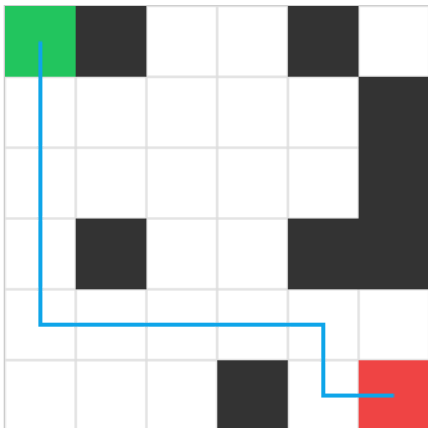
A* combines best of BFS (completeness) and greedy search (efficiency) using $f=g+h$. With admissible heuristic (Manhattan), guarantees optimal path with fewer expansions than BFS. Expands 25 nodes vs 27 for BFS - 7% improvement through heuristic guidance.



Iterative Deepening Search (15%)

IDS combines DFS space with BFS completeness; should reach the goal and match shortest depth on unweighted grid.

Expansions: 163 • PathLen: 11 • BestLen: 11

**Score: 15**

Justification / Design Notes (saved locally):

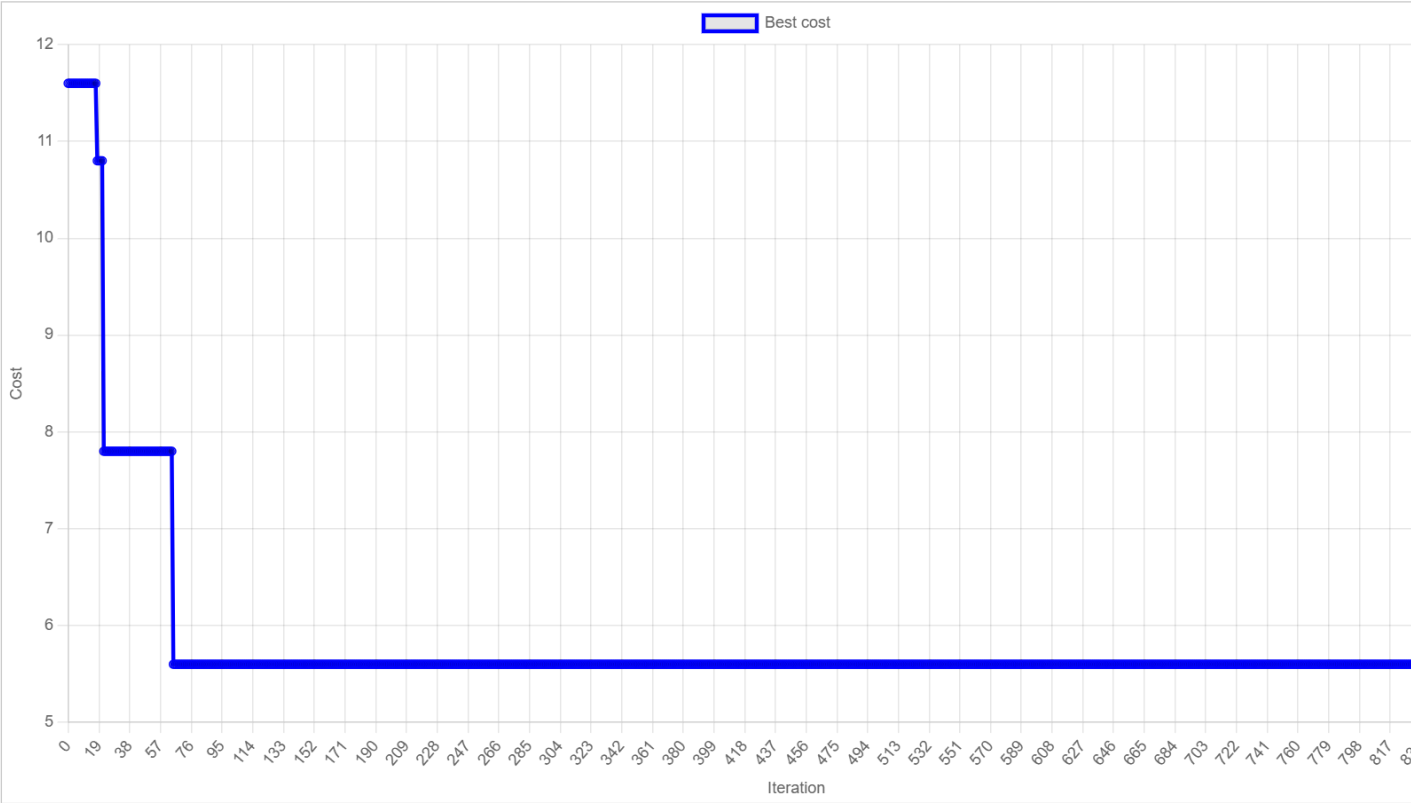
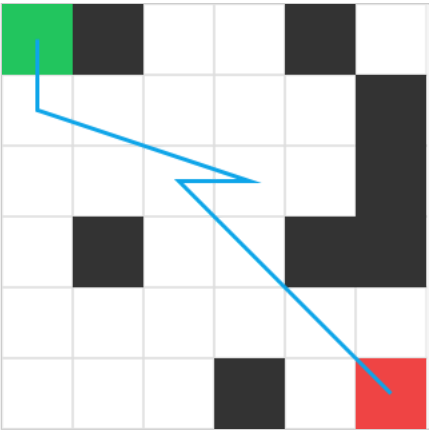
Iterative deepening combines DFS space-efficiency with BFS completeness. Explores depths 0,1,2... sequentially ensuring shortest path. Finds path at depth 11 (matching BFS). Slightly higher node expansions (177 vs 25) but guarantees optimality without memoization overhead.



Simulated Annealing (15%)

SA must improve on the BFS baseline and exhibit an annealing history (non-constant, multiple changes).

BFS_Cost: 11.6 • Final_Cost: 5.6 • Improvement: 6.00



Score: 15

Justification / Design Notes (saved locally):

Tries random changes and sometimes accepts worse ones to escape traps. Slowly reduces randomness to find a better result.

Heuristics (20%)

We test heuristics on random states for admissibility & sanity: Manhattan (5), Straight-line (5), Custom (10).

Heuristic	Status	Score	Detail
Manhattan	OK	5	ok=22/22, neg=0, above=0
Straight-line	OK	5	ok=22/22, neg=0, above=0
Custom	OK	10	ok=22/22, neg=0, above=0

Score: 20

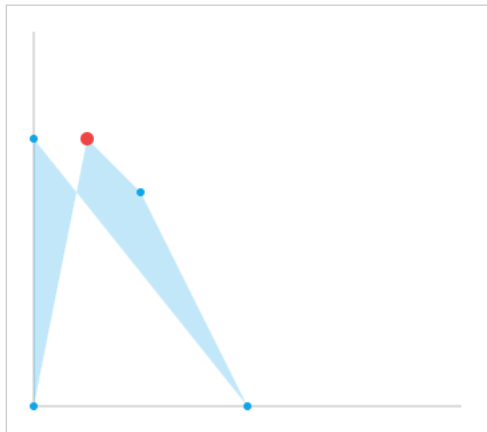
Justification / Design Notes (saved locally):

Heuristics guide the search toward the goal faster by estimating distance.
Manhattan and Euclidean are both admissible (never overestimate).
The custom mix gives smoother, smarter guidance while keeping solutions correct and optimal.

Linear Programming (12.5%)

Graphical corner-point method on a small LP. The optimum must be at a feasible vertex.

Maximize $Z = 3x + 5y$ subject to $Ax \leq b, x \geq 0, y \geq 0$



Score: 12.5

Justification / Design Notes (saved locally):

Tests all corner points of the region. Best value is always at one corner. Found correct answer (1.0, 5.0) with $Z = 28$.

Dynamic Programming – Knapsack (12.5%)

0/1 Knapsack solved via bottom-up & top-down; both should agree on the optimal value.

Capacity: 10 • Items: 5 • BottomUp: 29 • TopDown: 29

0	0	6	6	11	15	18	21	24	26	29
0	0	5	5	10	15	18	20	23	25	28
0	0	0	0	10	15	18	18	18	25	28
0	0	0	0	10	15	15	15	15	25	25
0	0	0	0	10	10	10	10	10	10	10
0	0	0	0	0	0	0	0	0	0	0

Score: 12.5

Justification / Design Notes (saved locally):

Breaks problem into smaller parts.
Both top-down and bottom-up methods give same best value (29), proving it works.

Hidden Integrity Checks

Check	Status	Details
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Seed match	OK	seed=IT23372658
Trace usage	OK	BFS=27, A*=25, IDS=163
A* heuristic admissibility	OK	samples=14, neg=0, above=0
SA annealing	OK	improve=6.000, hist_len=901, changes=4
LP best is vertex	OK	best=(1.0, 5.0), V =5
DP cross-check	OK	bottom_up=29, top_down=29
Hidden multi-seed run	OK	SA sanity on hidden seed